BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 109 · Lesson 109 · CPU Scheduling: FCFS, SJF and SRT

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Name the routine executed when an interrupt is accepted.

Q2. An OR gate has inputs 0 and 1. What is the output?

Q3. What is the effect of enqueue(X), enqueue(Y), dequeue() on an initially empty FIFO queue?

Q4. What is the time complexity of sequential search in the worst case?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

FCFS — Jobs run in first-arrival order.

SJF — Selects the shortest estimated ready job.

SRT — Preemptive shortest remaining time.

waiting time — Time spent ready but not executing.

New learning

FCFS — First come, first served: non-preemptive queue in arrival order.

SJF — Shortest job first: choose least burst time from ready jobs; non-preemptive.

SRT — Shortest remaining time: preempt if a new job has a shorter remaining burst.

Worked example

FCFS: P1 arrives at 0 (burst 4), P2 at 1 (burst 2), P3 at 2 (burst 1). Find waiting times.

1. Model answer — Timeline P1 0–4, P2 4–6, P3 6–7. Waiting = start − arrival: P1 0, P2 3, P3 4. Average = 7/3 ≈ 2.33 units. — Check: link each claim to the scenario and use precise OCR terminology.

We Do · Practise with guidance

SJF with all ready at 0: A burst 6, B burst 2, C burst 3. Trace and calculate mean waiting time.

What must we decide?

Identify the inputs, relevant facts and exact output or judgement required.

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

Order B 0–2, C 2–5, A 5–11. Waits B 0, C 2, A 5; mean 7/3 ≈ 2.33 units. SJF assumes burst-length estimates are available.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Distinguish non-preemptive and preemptive scheduling and give an example of each. [6]

Question 2

FCFS: A arrives 0/burst 3, B arrives 1/burst 2, C arrives 2/burst 4. Calculate each wait. [6]

Question 3

SRT: A arrives 0/burst 5, B arrives 1/burst 2. Show the timeline and explain preemption. [8]

Exit Ticket · Answer before checking

SRT never interrupts an executing process.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Interrupt service routine (ISR).

Q2. 1.

Q3. X is removed; Y remains at the front.

Q4. O(n).

You Do pages · Independent questions

Q1. Non-preemptive runs selected job until completion/block [2]; preemptive may interrupt running job [2]; correct algorithm examples [2].

Q2. A executes 0–3, wait 0; B 3–5, wait 2; C 5–9, wait 3. Two marks for correct timeline and four for wait calculations.

Q3. A 0–1; B 1–3; A 3–7 [6]. At t1 A has 4 remaining but B needs only 2, so B preempts [2].

Exit Ticket · Model answer

False. SRT is preemptive.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: